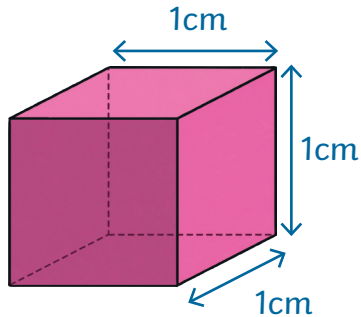


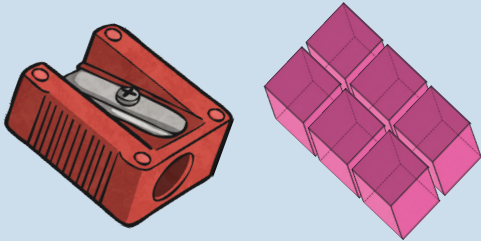
Measurement: Estimate Volume (Ages 9 – 10)

Volume is the amount of space a 3D object takes up.

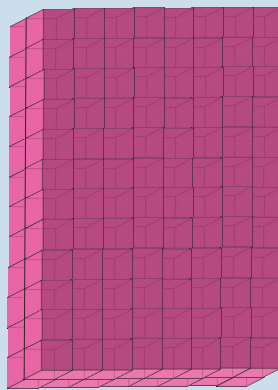


Here is a cube. Its volume is 1cm^3 .

We can estimate the volume of items using these cubes:



This pencil sharpener has an approximate volume of 6cm^3 .

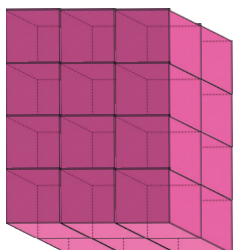


This notebook has an approximate volume of 192cm^3 .

Archie's Shapes

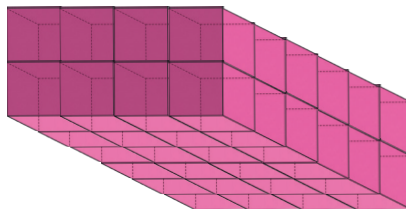
Archie makes some cuboids with his cubes. What is the volume of each cuboid?

1



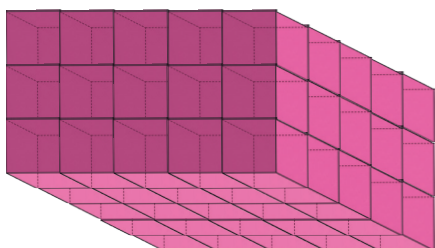
Volume = ____

2



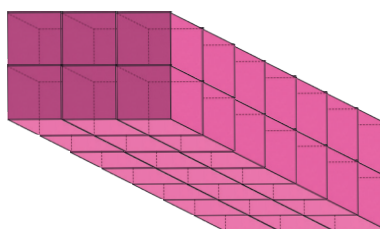
Volume = ____

3



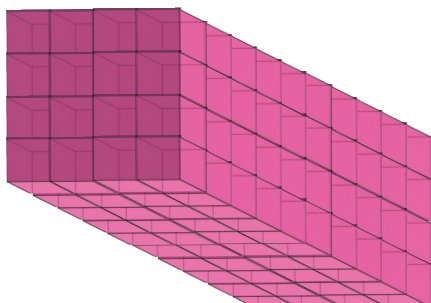
Volume = ____

4



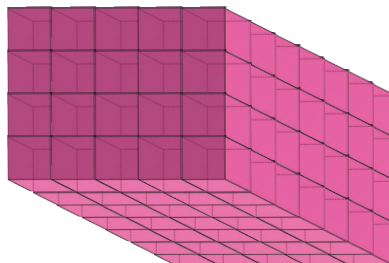
Volume = ____

5



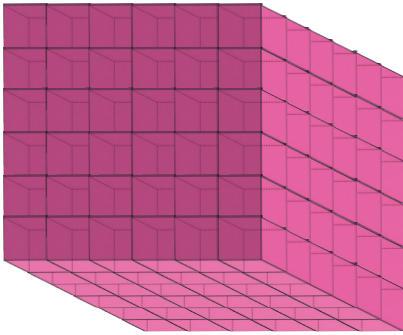
Volume = ____

6



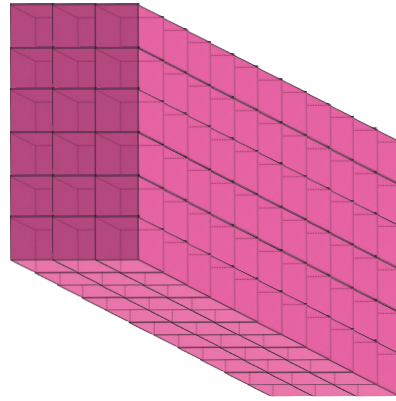
Volume = ____

7



Volume = ____

8



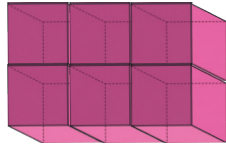
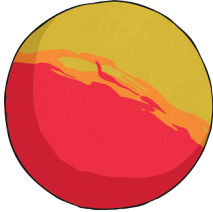
Volume = ____

Jen's Estimates

Jen uses her cubes to estimate the volume of a range of items.

Do you think her estimates are accurate? Decide whether you agree or whether you think her estimate is too big, too small or about right and give your reasons.

1

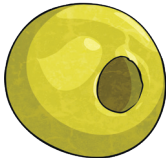


Jen says, "This bouncy ball is about 6cm^3 ".

Her estimate is...

too small.	about right.	too big.
Reason:		

2

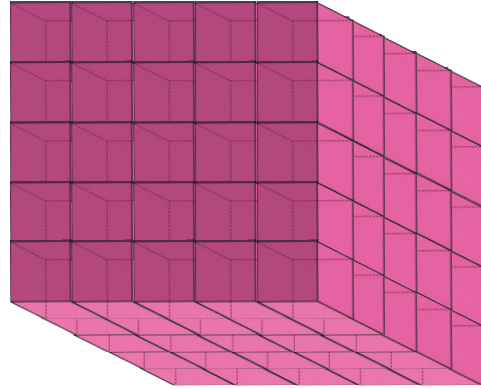
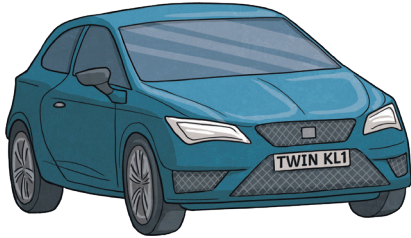


Jen says, "This olive is about 2cm^3 ".

Her estimate is...

too small.	about right.	too big.
Reason:		

3

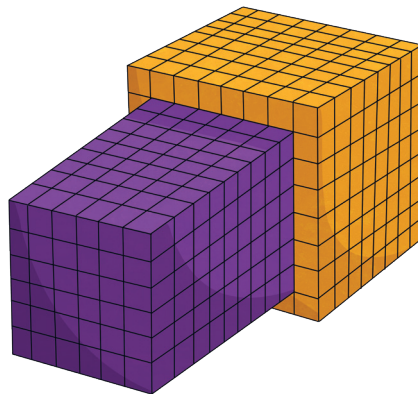
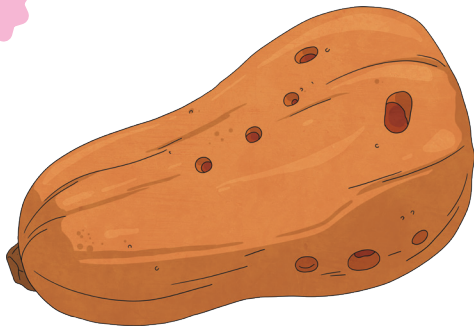


Jen says, "This toy car is about 125cm^3 ."

Her estimate is...

too small.	about right.	too big.
Reason:		

4



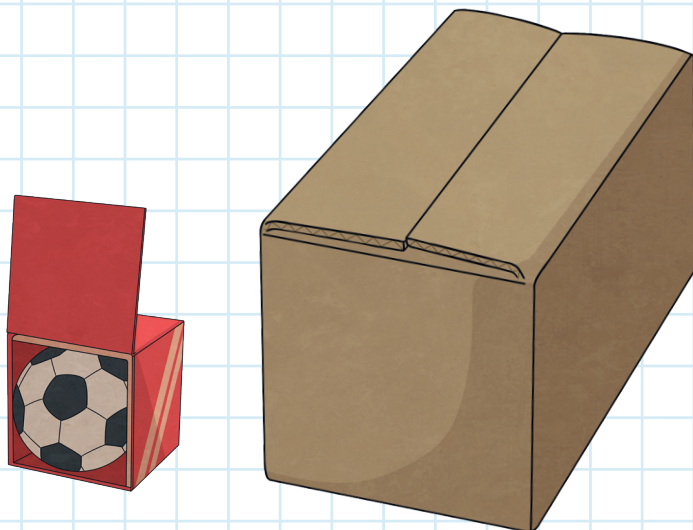
Jen says, "The volume of the butternut squash is 836cm^3 ."

Her estimate is...

too small.	about right.	too big.
Reason:		

More Estimates

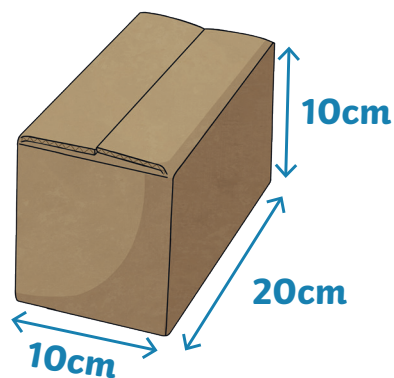
This football comes in a presentation box. A shop orders a large box of footballs. The shopkeeper says, "I think only 4 footballs will fit into this box."

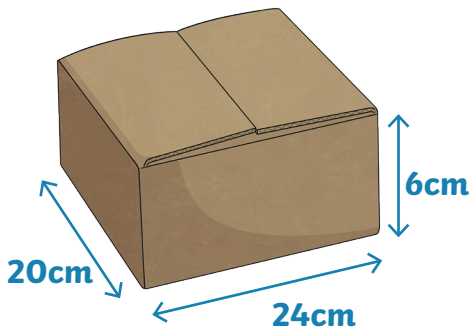
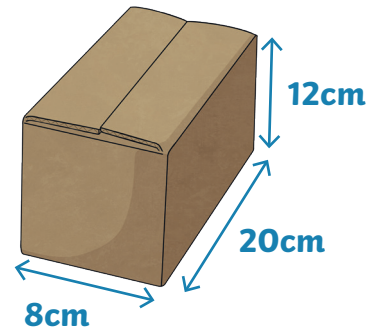
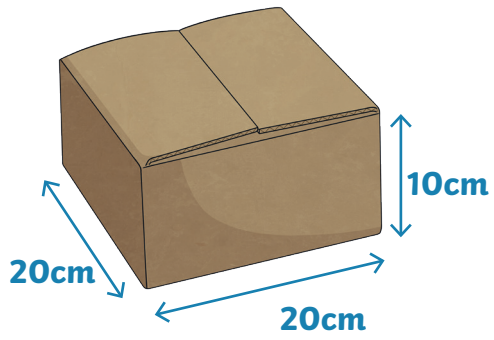


Explain why the shopkeeper is wrong.

Here are the dimensions for a pack of cards:

Decide whether 5 packs of cards could fit into each box and tick or cross it, then explain your reasons.





Measurement: Estimate Volume Answers

Archie's Shapes

1. Volume = $3\text{cm} \times 2\text{cm} \times 4\text{cm} = 24\text{cm}^3$
2. Volume = $4\text{cm} \times 6\text{cm} \times 2\text{cm} = 48\text{cm}^3$
3. Volume = $5\text{cm} \times 5\text{cm} \times 3\text{cm} = 75\text{cm}^3$
4. Volume = $7\text{cm} \times 3\text{cm} \times 2\text{cm} = 42\text{cm}^3$
5. Volume = $10\text{cm} \times 4\text{cm} \times 4\text{cm} = 160\text{cm}^3$
6. Volume = $7\text{cm} \times 5\text{cm} \times 4\text{cm} = 140\text{cm}^3$
7. Volume = $6\text{cm} \times 6\text{cm} \times 6\text{cm} = 216\text{cm}^3$
8. Volume = $11\text{cm} \times 3\text{cm} \times 6\text{cm} = 198\text{cm}^3$

Jen's Estimates

1

too small. ✓

about right.

too big.

Reason:

She hasn't taken into account the depth of the ball, so she will need another layer of cubes to be more accurate.

2

too small. ✓

about right.

too big.

Reason:

Two cubes together are not as big as the front surface of the olive and she has not added any layers to account for its depth.

3

too small.

about right.

too big. ✓

Reason:

She has made a large cube with the 1cm^3 cubes which is much larger than the whole toy car.

4

too small.

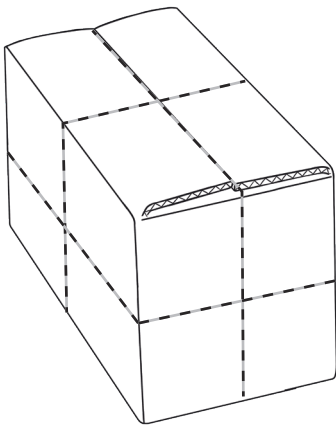
about right. ✓

too big.

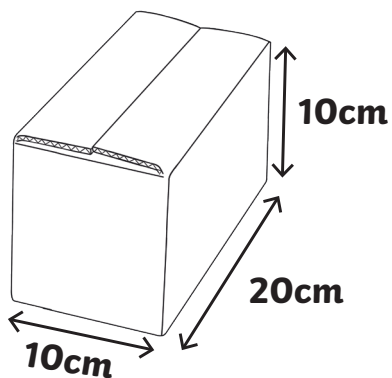
Reason:

She has replicated the simple shape and dimensions of the butternut squash. Although not completely accurate, it is a better estimate than one cuboid would give.

More Estimates

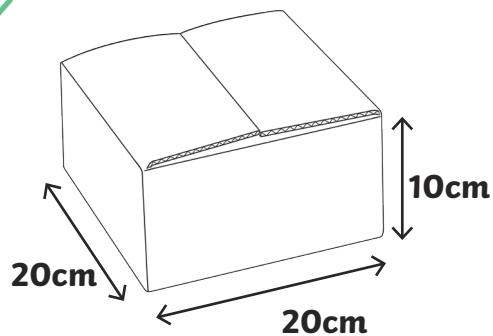


4 footballs could fit into the first layer of the box, so there is room for another layer.



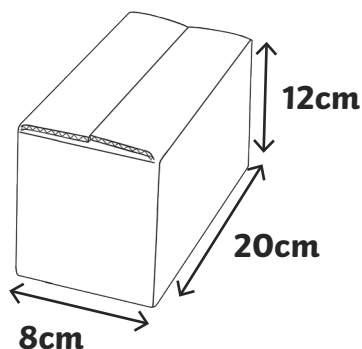
Reasons:

The box is big enough for the longest side of the pack of cards, but the next longest side is 12cm and the other two sides are only 10cm so the cards won't fit.



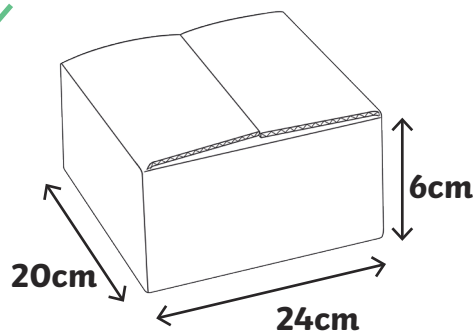
Reasons:

The bottom area of the box is 20cm x 20cm, so it would fit the pack of cards in, which is 20cm x 12cm. The height of the box is 10cm. If a pack of cards is laying down in the bottom of the box, it has a height of 2cm so 5 packs would have a height of 10cm, which will fit.



Reasons:

The box has the dimensions to fit a pack of cards in as the area of one side is 20cm x 12cm and the area of the largest side on the pack of cards is 20cm x 12cm. However, 5 packs of cards would have a depth of 10cm and the depth of this box is only 8cm.



Reasons:

The area of the bottom of the box is 24cm x 20cm. The area of one pack of cards is 20cm x 12cm, so 2 packs would fit on the bottom. The box has a height of 6cm, meaning up to 6 packs of cards could fit into this box.